Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Jin Kang	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University Dept. of electrical and computer engineering			
Address	3400 N. Charles St., Baltimore, MD 21218			

2. Technology Overview

Title of Proposed Technology	Endoscope-Holding Robotic System with Image-Guided Navigation for Single-Position Minimally Invasive Spine Surgery			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Mechanical Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Surgical Robots	Robotics/Automation Machinery	Robot Design Technologies
Concept of Proposed Technology	The proposed technology integrates an **endoscope-holding robotic system** with **image-guided navigation** to support **minimally invasive spine surgery**, including unilateral biportal endoscopy (UBE) and oblique lumbar interbody fusion (OLIF), performed in a **single patient position**. The platform is designed to improve surgical accuracy, reduce radiation exposure, and enhance workflow efficiency through the convergence of robotics, medical imaging, and navigation algorithms. **Core Technology Components** **1) Endoscope-Holding Robotic System** • Six-degree-of-freedom (6-DOF) collaborative robotic arm for endoscope positioning • Stable surgical field visualization without surgeon fatigue • Enables bimanual surgical operation by freeing both hands of the surgeon			

	2) CT–DRR Registration Algorithm • Matching of pre-operative CT images with digitally reconstructed radiographs (DRR) • Real-time 2D/3D registration using standard 2D C-arm fluoroscopy (no 3D C-arm required) • Target registration accuracy: ≤ 1.0 mm **3) Robotic Navigation System** • Calibration protocol linking robot and C-arm coordinate systems • Real-time trajectory guidance for cage and screw insertion • Drift and tremor compensation with correction rate greater than 90% **4) Single-Position Surgical Workflow** • Elimination of intraoperative patient repositioning • Reduction of total surgical time by more than 30% • Reduction of radiation exposure by more than 50% This integrated system enables precise navigation-guided spinal procedures while maintaining a cost-effective configuration based on standard imaging equipment.

Rationale for Joint R&D Project	Joint R&D collaboration is required due to the multidisciplinary complexity of the proposed system and the need for both algorithmic advancement and clinical validation.
	
	Key rationales include:
	
	1. **Algorithm Accuracy Enhancement**
	  ○ Integration of advanced expertise in computer-aided surgery to improve 2D/3D image registration accuracy and robustness
	2. **Clinical Validation**
	  ○ Access to diverse clinical datasets and standardized validation methodologies for comprehensive system verification
	3. **Global Market Entry Preparation**
	  ○ Joint intellectual property development and preparation for international regulatory pathways (e.g., CE and FDA certification)
	4. **Cost-Effective Surgical Navigation**
	  ○ Development of a navigation system based on standard 2D C-arm imaging to avoid reliance on expensive 3D imaging equipment, improving accessibility for medical institutions
	
	International joint R&D ensures scientific rigor, clinical relevance, and regulatory readiness while reducing development risks.

Key Words	1) Endoscopic Spine Surgery 2) Robotic Surgery 3) Image-Guided Navigation 4) CT–DRR Registration 5) Single-Position Surgery

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The objective of this project is to develop a **single-position endoscopic OLIF surgical workflow** integrating: • An endoscope-holding robotic system for stable visualization • Navigation-guided cage insertion without patient repositioning • A radiation-reduced surgical environment for medical staff			
R&D Target Specifications	**Quantitative Performance Targets:** • CT–DRR registration accuracy: ≤ 1.0 mm • Entry point offset: ≤ 1.5 mm • Angular offset: ≤ 2.5° • Radiation exposure reduction: ≥ 50% • Surgical time reduction: ≥ 30% • Cage insertion accuracy: ≥ 85% **System Specifications:** • Robot system: 6-DOF collaborative robot with medical-grade safety features and regulatory readiness • Navigation: Markerless CT-to-fluoroscopy registration with real-time tracking • Software: ROS-based robot control and real-time image processing pipeline • Validation pathway: Phantom testing followed by cadaver studies			
R&D Plan	**Year 1: System Development and Phantom Validation** • Development of endoscope-holding software and robotic integration • Establishment of robot–C-arm coordinate calibration protocol • Phantom-based accuracy testing (target entry offset < 1.5 mm) • Design and documentation of single-position surgical workflow **Year 2: Preclinical Validation and System Optimization** • Optimization of robotic system performance • Completion of integrated endoscope-navigation software • Cadaver studies for verification of accuracy and radiation reduction • Development of positioning guidance for safe surgical corridors **Year 3: Standardization and Commercialization Preparation** • Development of clinical practice guidelines (CPG) • Completion of surgical training materials (manuals, videos, protocols) • Patent applications covering workflow, control algorithms, and positioning guides • Technology transfer and commercialization preparation			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Enhance and validate 2D/3D image registration algorithms • Optimize surgical workflow through computational analysis • Design and conduct clinical validation studies for regulatory approval • Lead academic research and joint publications on safety and efficacy • Provide technical consultation in computer-aided surgery • Establish validation methodologies and quality assurance protocols
Expected R&R of Korean SME Partner	• Develop endoscope-holding robotic hardware optimized for spine surgery • Integrate robotic and navigation systems and perform system testing • Develop medical control algorithms and interface software • Coordinate phantom and cadaver testing with clinical partners • Establish medical device production and quality systems • Lead commercialization, market entry preparation, and sales strategy

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Establishment of a digital spine surgery standard using an endoscope-holding robotic system • Development of a cost-effective navigation system based on standard 2D C-arm imaging • Creation of an intellectual property portfolio including patents on workflow, control algorithms, and positioning guidance **Clinical Impact** • Reduction of radiation exposure by more than 50% • Reduction of total surgical time by more than 30% • Achievement of cage insertion accuracy greater than 85% • Improvement of surgeon safety through near-zero radiation exposure **Economic and Industrial Impact** • Entry into the global spine surgery robotics market • Standardization of surgical training and education packages for clinical adoption • Strengthening of competitiveness in medical robotics and navigation technologies **Academic Outcomes** • Publication of joint research papers in high-impact journals • Development of standardized clinical practice guidelines for robotic spine surgery

6. References

(1) Strategic Market Research. "Spine Surgery Robots Market Size Report 2024-2030." Available at: https://www.strategicmarketresearch.com/market-report/spine-surgery-robots-market

(2) Persistence Market Research. "Spine Surgery Robots Market Trends & Forecast to 2031." Available at: https://www.persistencemarketresearch.com/market-research/spine-surgery-robots-market.asp